Review plan · for the lecturer

Teaching programming and algorithms at university: unit check

Programming · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Asymptotic analysis
- Recursion and induction
- Notional machines
- Assessment at scale

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change your assignments?

Exit ticket

- One change to my programming course this term, and how I will know it worked.

Try it in your next lecture
Bring one redesigned assignment, its hidden tests and your AI policy to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **$5n^3 + 2n$ is**

◯ $\Theta(n^5)$ ◯ $\Theta(n^3)$ ◯ $\Theta(n)$

2 **Solve the recurrence.**

$T(n) = 2T(n/2) + 1$ gives $\Theta($ _____ $)$.

Word bank: n log n · n · log n

3 **Big-O gives an upper bound on how running time grows, not a lower one.**

◯ True ◯ False

4 **Fact(n) with no reachable base case ends in**

◯ A stack overflow ◯ Returning 0 ◯ A syntax error

5 **Order the induction proof.**

☐ Inductive step

☐ Conclusion

☐ Base case

☐ Inductive hypothesis

6 **What does this program print?**

```
x = 1
y = x
x = 9
print(y)
```

◯ 1 and 9 ◯ 9 ◯ 1

7 **Which reveal a student's mental model? Choose all that apply.**

☐ Explain-in-plain-English questions ☐ Tracing questions ☐ Whether the submitted code passes all the tests

8 **Match each tool to its use.**

Hidden tests	• • • •	Measure the suite
Mutation testing	• • • •	Flag similar code
MOSS	• • • •	Catch edge cases

9 **AI-text detectors are reliable enough to prove that code was machine-written.**

◯ True ◯ False

10 **Best evidence a student understands submitted code:**

◯ Modifying and explaining it live ◯ A low MOSS similarity score ◯ A perfect autograder score

Answers · for the lecturer

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1 **$\Theta(n^3)$**

Constants and lower terms vanish.

2 **T(n) = 2T(n/2) + 1 gives $\Theta(n)$.**

n^(log₂2) = n dominates f(n) = 1.

3 **True**

It is an upper bound; Ω is the lower bound, and Θ holds when both do.

4 **A stack overflow**

Frames pile up without end.

5 **Top to bottom: 3, 4, 1, 2**

Base, assume, step, conclude.

6 **1**

y refers to the value x had then (1); x = 9 makes only x refer to a new value.

7 **Tracing questions; Explain-in-plain-English questions**

Passing tests can come from pattern-matching or from AI; tracing and explaining show the model.

8 **Hidden tests — Catch edge cases**
Mutation testing — Measure the suite
MOSS — Flag similar code

Each answers a different question.

9 **False**

They are unreliable; design the assessment instead.

10 **Modifying and explaining it live**

A score shows what the code does or that it is unlike others' code, not that the student understands it.